

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: OE& NA**

2. **Subject Name: Seakeeping L-T-P: 3-0-0 Credits:3**

3. **Pre-requisites:**

4. **Syllabus and reference books:**

Syllabus:

Introduction to sea-keeping. Recapitulation of surface gravity waves. Ship motions in regular waves: Frequency of encounter. Uncoupled Heaving, pitching and rolling motion. Natural periods; Froude Scaling; Coupled heaving and pitching problem, Wave loads. RAO. Description of irregular waves: wave spectrum. Motions in irregular waves: response spectra. Derived responses: slamming, deck wetness, relative motions, sea-sickness etc. Motion stabilization. Model Tests, Full-Scale trials and scale effect. Design considerations for seakeeping.

Reference Books:

- 1) Dynamics of Marine Vehicles by Rameswar Bhattacharyya
- 2) Sea Loads on Ship and Offshore Structures by O M Faltinsen
- 3) Principles of Naval Architecture, Part 3 by SNAME Principle

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Introduction to sea-keeping	2
2.	surface gravity waves	4
3.	Ship motions in regular waves	4
4.	Frequency of encounter	3
5.	Uncoupled Heaving, pitching and rolling motion	5
6.	Coupled heaving and pitching problem	5
7.	Wave loads. RAO	3
8.	irregular waves: wave spectrum	3
9.	Motions in irregular waves: response spectra	3
10.	Derived responses	4
Total number of hours		36

